PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Technology-Based Interventions for Nursing Home Residents: A Systematic Review Protocol
AUTHORS	Su, Zhaohui; Meyer, Kylie; Li, Yue; McDonnell, Dean; Joseph, Nitha; Li, Xiaoshan; Du, Yan; Advani, Shailesh; Cheshmehzangi, Ali; Ahmad, Junaid; da Veiga, Claudimar; Chung, Roger; Wang, Jing; Du, Hongwu

VERSION 1 – REVIEW

REVIEWER	Milica Petrovic Catholic University of the Sacred Heart, Psychology
REVIEW RETURNED	13-Aug-2021

GENERAL COMMENTS	This paper covers the topic of technology-based interventions for nursing home residents. The paper is clear and very well written, adhering to the PRISMA checklist. However, there are some aspects of the paper that I found confusing and in my opinion, attending to those aspects could further improve the paper. For instance, in the abstract, the authors note "This systematic review will identify technology-based interventions designed for nursing home residents and describe the characteristics and effects of these interventions concerning/span> the distinctive traits of nursing home residents and nursing facilities" making it a bit difficult to understand what "concerning/span> the distinctive traits" actually means. Furthermore, the text continues: "Additionally, this paper will present practical insights into the varying intervention approaches that can assist in the delivery of broad digital health solutions for nursing home residents amid and beyond the impact of COVID-19". Does this mean that one of the aims is to explore how the intervention approach reflects on the success of the dissemination of digital solutions? In that case, it is confusing if the main focus of this paper is technology-based interventions or digital solutions. The background of the paper is very well written and includes extensive literature, reflecting the hard work of the authors in thoroughly exploring the topic, well done. I would advise a clearer distinction between technology-based interventions and technology-based solutions, elaborating on what constitutes an intervention and how can a digital tool be defined as an intervention tool. For instance, if the content provided on the mobile phone delivers education about a concrete illness or shares practical information in audio or video format, is it still considered an intervention? Moreover, I advise the word intervention should be used for evidence-based treatments only. I was able to partially see this distinction in the Inclusion/Exclusion criteria section but it was

	still not clear to me if the focus of this paper is on existing digital tools for nurse home residents or evidence-based treatments. Based on this sentence "Additionally, this research will present practical insights into the varying intervention approaches that can assist in the delivery of broad digital health solutions for nursing home residents amid and beyond the impact of COVID-19" two questions arise: Are there already pre-defined/existing intervention approaches (or even a list of such for digital interventions) that can be used as a template to compare your findings against? If not, could you only clarify what do you suggest when saying intervention approaches (e.g., ...) Is there any supporting literature that addresses the potential of an intervention approach to affect the dissemination of the digital solution or do you imply this? Within the "Study quality assessment" section or maybe "Ethics and dissemination" I advise authors to mention that studies included in the systematic review adhered to the ethical standards or received ethical approval.

REVIEWER	Antonios Tsertsidis Orebro University
REVIEW RETURNED	16-Aug-2021

GENERAL COMMENTS	The authors have submitted an interesting protocol paper. I believe that the background is provides a good motivation as to why this study should be conducted. The methods section is well described and as this will be a systematic review, the authors will adhere to the PRISMA guidelines. Overall, I think that the study is well planned. However, before submitting the full paper, some proof-reading should be considered.

VERSION 1 – AUTHOR RESPONSE

Responses to Reviewer No.1's Comments	
This paper covers the topic of technology-based interventions for nursing home residents. The paper is clear and very well written, adhering to the PRISMA checklist.	Thank you very much for your kind words and constructive input. We really appreciate you investing your time in helping improve this paper.
However, there are some aspects of the paper that I found confusing and in my opinion, attending to those aspects could further improve the paper. For instance, in the abstract, the authors note "This systematic review will identify technology-based interventions designed for nursing home residents and describe the characteristics and effects of these interventions concerning/span> the distinctive traits of nursing home residents and nursing facilities" making it a bit difficult to understand what "concerning/span>	Per the reviewer's suggestion, we edited the manuscript multiple times to eliminate elements that may cause confusion, including but not limited to the sections the reviewer has kindly pointed out.

the distinctive traits" actually means.	
Furthermore, the text continues: "Additionally, this paper will present practical insights into the varying intervention approaches that can assist in the delivery of broad digital health solutions for nursing home residents amid and beyond the impact of COVID-19". Does this mean that one of the aims is to explore how the intervention approach reflects on the success of the dissemination of digital solutions? In that case, it is confusing if the main focus of this paper is technology-based interventions or digital solutions.	Thank you very much for your comments. We thoroughly revised this section to remove the confusing parts (p.7). We also clarified the differences between technology-based interventions and concepts like digital solutions (p.9). We hope the revised manuscript meets the reviewer's expectations.
The background of the paper is very well written and includes extensive literature, reflecting the hard work of the authors in thoroughly exploring the topic, well done.	Thank you very much for your kind words. They mean a great deal to us.
I would advise a clearer distinction between technology-based interventions and technology-based solutions, elaborating on what constitutes an intervention and how can a digital tool be defined as an intervention tool.	We clarified the differences between these two concepts per the reviewer's advice (p.9). Thank you very much for your instructive input.
For instance, if the content provided on the mobile phone delivers education about a concrete illness or shares practical information in audio or video format, is it still considered an intervention? Moreover, I advise the word intervention should be used for evidence-based treatments only. I was able to partially see this distinction in the Inclusion/Exclusion criteria section but it was still not clear to me if the focus of this paper is on existing digital tools for nurse home residents or evidence-based treatments.	We agree with the reviewer–whether the educational materials are evidence-based is the key difference that separates the two concepts. Per the reviewer's advice, we provided a thorough explanation of the two concepts in the manuscript and revised the Inclusion/Exclusion criteria to reflect the change.
Based on this sentence "Additionally, this research will present practical insights into the varying intervention approaches that can assist in the delivery of broad digital health solutions for nursing home residents amid and beyond the impact of COVID-19" two questions arise: Are there already pre-defined/existing intervention approaches (or even a list of such for digital interventions) that can be used as a template to compare your findings against? If not, could you only clarify what do you suggest when saying intervention approaches (e.g., ...)	We agree with the reviewer—this section is confusing. We revised the parts that are in line with our research objectives and deleted the rest. Thank you very much for your kind help.

Is there any supporting literature that addresses the potential of an intervention approach to affect the dissemination of the digital solution or do you imply this?	We searched the literature and could not find evidence that is relevant to our research context. We do think this calls for systematic investigation—how the interventions are delivered (e.g., whether they require nursing home residents' input, like daily login/data input, compared to those that are sensor-based) may have an impact on outcomes such as user engagement.
Within the "Study quality assessment" section or maybe "Ethics and dissemination" I advise authors to mention that studies included in the systematic review adhered to the ethical standards or received ethical approval.	We added this section per the reviewer's suggestion. Thank you very much for your comments.
Responses to Reviewer No.2's Comments	
The authors have submitted an interesting protocol paper. I believe that the background is provides a good motivation as to why this study should be conducted. The methods section is well described and as this will be a systematic review, the authors will adhere to the PRISMA guidelines. Overall, I think that the study is well planned. However, before submitting the full paper, some proof-reading should be considered.	Thank you so much for your kind words and constructive input—We are beyond grateful. Following your suggestions, we edited the manuscript multiple rounds. We hope the updated paper meets your expectations, and thank you very much for helping us improve this research.

VERSION 2 – REVIEW

REVIEWER	Milica Petrovic Catholic University of the Sacred Heart, Psychology
REVIEW RETURNED	06-Oct-2021

GENERAL COMMENTS	This is a much-improved version of the manuscript. I would like to thank the authors for addressing the comments so thoroughly and hope they will keep up the good work.